Electronic PULSE WAVEFORM MONITOR

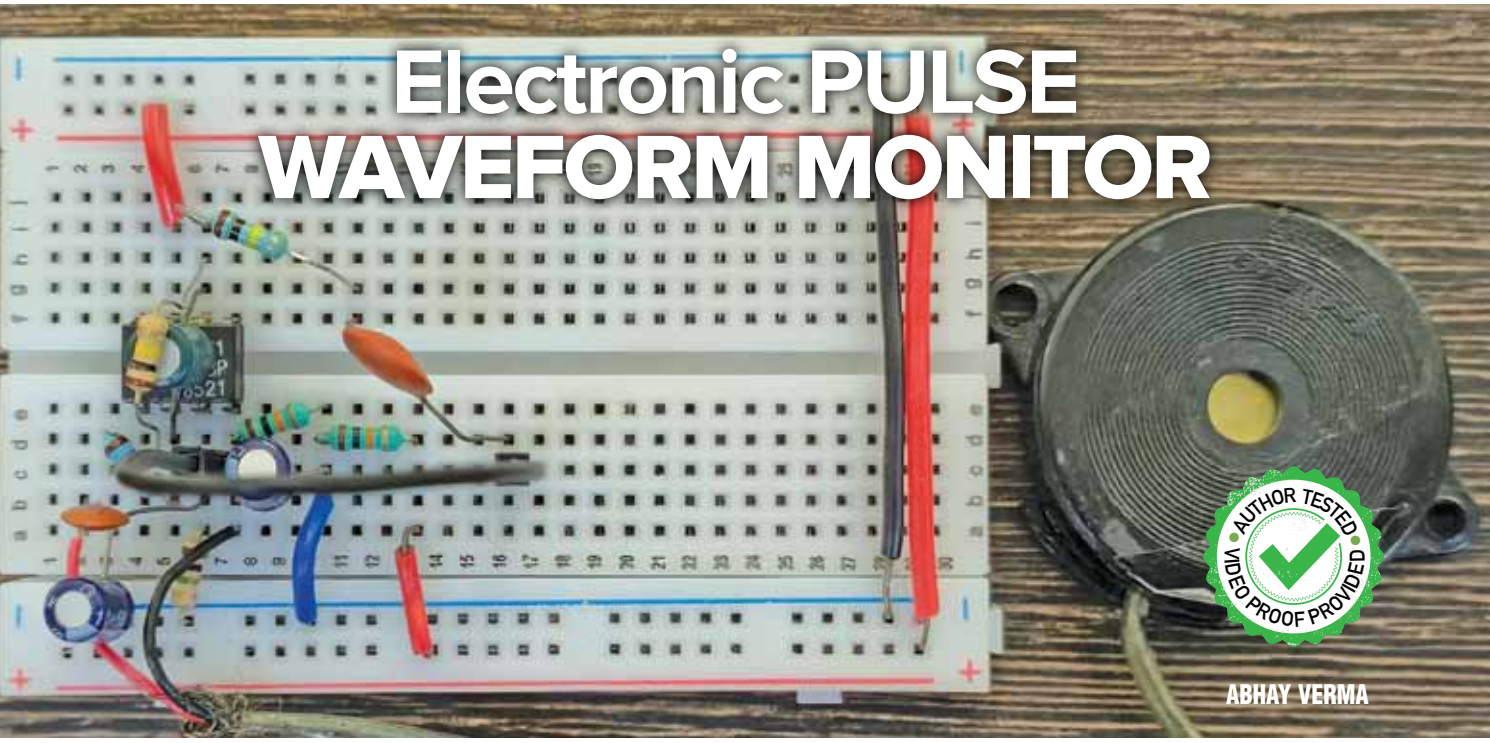


Fig. 1: Author's prototype

The electronic pulse waveform monitor demonstrates low-frequency signal amplification, high-impedance matching, and analogue signal processing. Designed to operate from a low-voltage DC supply, the device captures small mechanical vibrations produced by the body and converts them into a high-gain, conditioned analogue voltage waveform. Unlike standard airborne acoustic microphones, its contact-based piezoelectric transducer detects direct body vibrations, allowing physiological pulses to be displayed in real time on a digital storage oscilloscope (DSO). Fig. 1 shows the prototype on a breadboard.

The JFET-input TL071 operational amplifier is used to provide high-gain, low-noise amplification of the faint, low-frequency signals generated by the piezo sensor. A high-value resistor at the non-inverting input maintains the high input impedance needed to capture the weak sensor signal, while active and passive RC low-pass filters attenuate unwanted high-frequency noise. The circuit is assembled on a breadboard, with attention to supply decoupling, virtual-ground stability, and proper grounding to minimise mains interference.

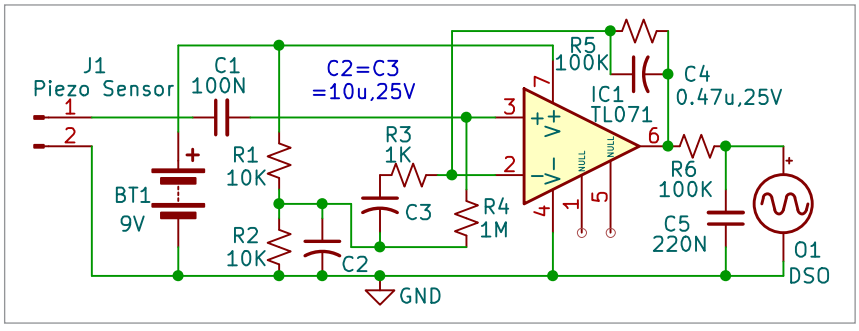


Fig. 2: Circuit diagram

Circuit and working

Fig. 2 shows the circuit diagram of the electronic pulse waveform monitor. It is built around a JFET-input operational amplifier TL071 (IC1), along with passive components for biasing, amplification, filtering, and signal conditioning. A 9V DC battery (BT1) provides a stable power supply, reducing interference from AC mains. Since the TL071 operates from a single-ended 9V supply, a 4.5V virtual-ground reference is created using two equal-value resistors, R1 and R2 (10kΩ each). Capacitor C2 (10µF) stabilises this reference against voltage fluctuations.

Abhay Verma is an electronics engineer at EFY and a passionate electronics hobbyist with a keen interest in practical circuit design and DIY projects.

Ready to bring this circuit to life? To unlock the circuit diagram, component list, PCB layout and complete step-by-step build instructions visit: https://efy.co.in/ebazb3